

ME8930: Data-driven Learning and Control of Dynamic Systems

Spring 2026

Final Project Timeline and Deliverables

1. **Project topic approval — due end of February.**
 - (a) Submit a 1-page proposal with title and team members
2. **Preliminary project report — due March 23.**

Include:

 - (a) Problem statement
 - (b) Literature review (key papers and positioning)
 - (c) Proposed approach and methodology
 - (d) Evaluation plan: dataset/platform, baselines, metrics
 - (e) Anticipated results and risk/mitigation plan
 - (f) Clear breakdown of tasks assignment and responsibilities of each member
3. **Progress report — due April 13.**

Add to the preliminary report:

 - (a) Preliminary results (plots/tables)
 - (b) Debugging lessons / revised plan
 - (c) Updated timeline to completion
4. **Final report and presentation — due April 29.**
 - (a) Final report (complete results, comparisons, and conclusions)
 - (b) Presentation (and demo when applicable). Use the IEEE conference style formatting from Overleaf (<https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhnncsfqn>)

Report Format Guidelines

1. Preliminary report: $\sim$ 3–5 pages (excluding references) + pdf of a presentation (3–5 slides) summarizing the proposed work.
2. Progress report: $\sim$ 5–7 pages
3. Final report: $\sim$ 8–12 pages including references (appendix optional) + code outcome as a zip file + public GitHub

Use clear figures, labeled axes, and short captions that state what each figure shows.

Presentation Guidelines

1. Time: 12 minutes talk + 3 minutes Q&A (adjustable depending on number of teams)
2. Required content:
 - (a) Problem and why it matters
 - (b) Overview of your proposed method
 - (c) Baseline methods for comparison and evaluation metrics used
 - (d) Results (including at least one failure case or limitation)
 - (e) Takeaways and future work

Grading Rubric

1. Preliminary report (problem, related work, plan quality):
2. Progress report (evidence of progress, preliminary results):
3. Final report (technical quality, evaluation, clarity):
4. Presentation (communication, insight, Q&A):

The preliminary report (50 %) and progress report (50 %) combined will count 30% of the final grades. The final report (70 %) and the presentation (30 %) will count 40 % of your final grades.

Collaboration and Academic Integrity

1. Teams may discuss ideas with other teams, but code, writing, and final results must be original. Ideally, two student teams will be preferred. Please discuss with me if you want to form a single or a three-student team.
2. Cite any external code, datasets, or libraries used.
3. Use AI tools responsibly: it may be used for brainstorming and debugging, but you must understand and be able to explain all submitted content.

Instructor Support

1. Teams are encouraged to meet briefly with the instructor before the submission of the preliminary and project report.
2. Office hours/meetings can be scheduled immediately after the class or any time based on mutual agreement.